AP BIOLOGY TOPIC 7.10

Speciation: Reproductive Isolation, Allopatric vs. Sympatric & Macroevolution

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-3.D | EVO-3.E | EVO-3.F | PRE/POSTZYGOTIC BARRIERS

Credits: mitali120501@gmail.com

§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.10 (SPECIATION: REPRODUCTIVE ISOLATION, ALLOPATRIC VS. SYMPATRIC & MACROEVOLUTION)

CED Core Vocabulary

Speciation — The evolutionary process by which one ancestral species splits into two or more reproductively isolated descendant species.

Reproductive Isolation Is Key — Prevents gene flow between two populations, allowing their gene pools to diverge independently via mutation, natural selection, and genetic drift.

Sympatric Speciation ("Same Country") — Occurs without geographic isolation while populations live in the exact same area, driven by: 1. Polyploidy (Instant Chromosomal Error in Meiosis):...

3. Sexual Selection — Female Lake Victoria cichlid fish selecting males of specific colors under different light wavelengths.

Biological Species Concept (Ernst Mayr) — Defines a species as a group of populations whose members have the potential to interbreed in nature and produce viable, fertile offspring , and cannot produce...

Allopatric Speciation ("Other Country") — Caused by geographic isolation (mountain range, canyon, river, isthmus, island colonization) physically separating a population and blocking gene flow. Examples:...

2. Habitat / Host Shift Differentiation — Apple maggot flies (Rhagoletis pomonella) shifting from native hawthorn trees to introduced apple trees.

Phyletic Gradualism — Evolution proceeds slowly and continuously as small adaptive changes accumulate over millions of years.

§2. 1. BIOLOGICAL SPECIES CONCEPT & REPRODUCTIVE ISOLATION

Definition of Species

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

1. Biological Species Concept & Reproductive Isolation ➔ **Speciation** ➔ **Biological Species Concept (Ernst Mayr)** ➔ **Reproductive Isolation Is Key**

- **Speciation:** The evolutionary process by which one ancestral species splits into two or more reproductively isolated descendant species.
- **Biological Species Concept (Ernst Mayr):** Defines a species as a group of populations whose members have the potential to **interbreed in nature and produce viable, fertile offspring**, and cannot produce viable, fertile offspring with other such groups.
  - *Limitations:* Does not apply to **asexual prokaryotes**, **extinct fossils**, or ring species (where morphological/phylogenetic/ecological species concepts are used instead).
- **Reproductive Isolation Is Key:** Prevents **gene flow** between two populations, allowing their gene pools to diverge independently via mutation, natural selection, and genetic drift.

[MECHANISTIC NOTE] **No gene flow = populations diverge genetically into separate species.**

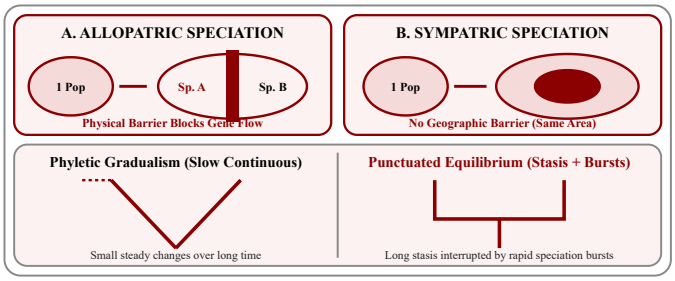
[AP EXAM FRQ SCORING STRATEGY] **3-Step FRQ Template for "How Do Two Populations Become Separate Species?":**
(1) Gene flow stops (geographic barrier or sympatric shift); **(2) Different selective pressures, random mutations, and genetic drift** act independently on each isolated population; **(3) Enough genetic divergence accumulates** to establish prezygotic or postzygotic reproductive isolation so they can no longer interbreed!

§4. 3. ALLOPATRIC VS. SYMPATRIC SPECIATION

With vs. Without Geography

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

3. Allopatric vs. Sympatric Speciation ➔ **Allopatric Speciation ("Other Country")** ➔ **Sympatric Speciation ("Same Country")**



- **Allopatric Speciation ("Other Country"):** Caused by **geographic isolation** (mountain range, canyon, river, isthmus, island colonization) physically separating a population and blocking gene flow.
  - *Examples:* Kaibab vs. Abert's squirrels on opposite rims of the Grand Canyon; snapping shrimp across the Isthmus of Panama.
- **Sympatric Speciation ("Same Country"):** Occurs **without geographic isolation** while populations live in the exact same area, driven by:
  - **1. Polyploidy (Instant Chromosomal Error in Meiosis):** Nondisjunction produces extra chromosome sets (**3n, 4n**) — **Autopolyploidy** (within 1 species) or **Allopolyploidy** (hybrid of 2 species). A tetraploid (**4n**) plant is instantly reproductively isolated from its diploid (**2n**) parent because **4n × 2n → sterile 3n triploid** offspring!
  - **2. Habitat / Host Shift Differentiation:** Apple maggot flies (*Rhagoletis pomonella*) shifting from native hawthorn trees to introduced apple trees.
  - **3. Sexual Selection:** Female Lake Victoria cichlid fish selecting males of specific colors under different light wavelengths.

[COMMON AP EXAM FRQ TRAP] **Polyploidy is an extremely common mechanism of sympatric speciation in PLANTS (e.g., wheat, cotton, strawberries), NOT most animals!**

§3. 2. PREZYGOTIC VS. POSTZYGOTIC REPRODUCTIVE BARRIERS

8 Isolation Mechanisms

BARRIER TYPE	SPECIFIC MECHANISM	CLASSIC BIOLOGICAL EXAMPLE
PREZYGOTIC *(Before zygote forms; prevents mating or fertilization)*	**1. Habitat Isolation**	Live in different microhabitats in same region (e.g., water-dwelling vs. terrestrial garter snakes; ground vs. tree frogs).
	2. Temporal Isolation	Breed at different times of day, seasons, or years (e.g., Eastern vs. Western spotted skunks breed in late winter vs. late summer).
	3. Behavioral Isolation	Incompatible courtship displays, songs, dances, or pheromones (e.g., blue-footed booby high-step dance; firefly flash patterns).
	4. Mechanical Isolation	Anatomically incompatible reproductive structures/genitalia or flower shapes (e.g., snail shell coil chirality; *Salvia* flowers).
	5. Gametic Isolation	Sperm cannot recognize or fuse with egg surface receptors (e.g., red vs. purple sea urchin bindin proteins in broadcast spawning).
POSTZYGOTIC *(After fertilization; prevents hybrid from becoming viable/fertile adult)*	**6. Reduced Hybrid Viability**	Hybrid zygote fails to develop or dies frail before maturity (e.g., *Ensatina* salamander subspecies hybrids).
	7. Reduced Hybrid Fertility	Hybrid grows into a healthy adult but is **sterile** due to uneven chromosome pairing in meiosis (e.g., **Mule** [**2n=63**] = Horse [**2n=64**] × Donkey [**2n=62**]).
	8. Hybrid Breakdown	First-generation (**F_1**) hybrids are viable & fertile, but **F_2** offspring are weak or sterile (e.g., cultivated rice strains).

§5. 4. PACE OF EVOLUTION & MACROEVOLUTIONARY PATTERNS

Gradualism vs. Punctuated

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

4. Pace of Evolution & Macroevolutionary Patterns ➔
Divergent Evolution & Adaptive Radiation ➔ **Convergent Evolution**

- **Gradualism vs. Punctuated Equilibrium (Eldredge & Gould):**
  - **Phyletic Gradualism:** Evolution proceeds slowly and continuously as small adaptive changes accumulate over millions of years.
  - **Punctuated Equilibrium:** Species experience long geological periods of morphological **stasis (equilibrium)** punctuated by rapid bursts of speciation when environments change sharply.
- **Divergent Evolution & Adaptive Radiation:**
  - **Divergent Evolution:** Closely related populations from a common ancestor accumulate differences to adapt to different habitats.
  - **Adaptive Radiation:** Rapid evolution of many diverse species from a **single ancestral species** when novel ecological niches open up (e.g., **Darwin's Galápagos finches**, Hawaiian silverswords, Cenozoic mammals after dinosaur extinction).
- **Convergent Evolution:** Unrelated species evolve similar analogous traits under similar environmental pressures (does *not* indicate close ancestry!).

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§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.10 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.10: Speciation: Reproductive Isolation, Allopatric vs. Sympatric & Macroevolution ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + $2pq$ (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate $2pq$ (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.10 to prove why the claim holds.

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